Transducers & Sensors :-

Sensor

Sensor is a device that detect a physical qu... (press., temp., disp.) and convert it into a mea... signal or unit.

Transducer :

It is a device which converts one form of ene... into another form.

Note :- All sensors are transducer but all transdu... are not be sensor.

Classification of transducer :-

(1) On the basis of energy source :-

- Active Transducer :- These type of transducer doesn't require any ext. power source
  Eg thermocouple, piezo. electric sensor, photo voltanic cell.
- Passive Transducer :- These type of transducer requi... ext. power source
  Eg strain gauge, LVDT (Linear variable differential tra...

Q. based on strain gauge:-

$$\epsilon \propto \frac{\Delta R}{R} \Rightarrow \epsilon = (G.F) \times \frac{\Delta R}{R}$$

[R = Resistance, G.F = Gauge factor, ϵ = Strain]

Qn A strain gauge has G.F = 2, original resistance (R) = 120Ω. Strain applied (ϵ) = 500×10^{-6}. Find the change in resistance.

Soln $\epsilon = (G.F) \times \frac{\Delta R}{R}$

$$500 \times 10^{-6} = 2 \times \frac{\Delta R}{120}$$

$$\frac{500 \times 120 \times 10^{-6}}{2} = \Delta R \Rightarrow \Delta R = 3 \times 10^{-2}$$

$$\boxed{\Delta R = 0.03}$$

Change = final - initial

Qn Based on ~~thermocouple~~ thermister @ resistance temp. detector (RTD)

$R(T) = R_0(1 + \alpha T)$ at 0°C

Qn An RTD has $R_0 = 100\Omega$ at 0°C, coeff. of thermal expansion (α) = 0.004/°C determine the value of R at 50°C.

Soln $R = R_0(1 + \alpha T)$

$$R = 100\left(1 + \frac{0.004}{1000} \times 50^\circ\right)$$

$$= 100\,(1 + 0.2)$$

$$\boxed{R = 120\Omega}$$

Wheatstone bridge :- (Quarter bridge)

$$V_f = \frac{V_i}{4} \times G.F \times \epsilon$$

Qu. A strain gauge is used in wheatstone bridge. The given datas are Supply voltage (V) = 5 V, gauge factor (G.F) = 2, $\epsilon = 1000 \times 10^{-6}$, determine final voltage

$$V_f = \frac{5}{4} \times 2 \times 1000 \times 10^{-6}$$

$$= \frac{100}{4} \times 10^{-4}$$

$$= 25 \times 10^{-4}$$

$$\boxed{V_f = 2.5 \text{ mV}}$$

(2) On the basis of O/P
- Analog
- Digital

Analog :- O/P is continous
Eg thermister, LVDT

Digital :- O/P is decrete in the form of 0 & 1
Eg Optical encode.

① Based on the principle of operation:-

	Principle	Ex.
(1)	Resistive	Strain gauge, thermister, RTD
(2)	Inductive	LVDT
(3)	Capacitive	Capacitive displ. Sensor
	piezo electri	force sensor
	optical	photo diode
(6)	Hall effect	~~magnet~~ $\vec{B}$ sensor

② Based on measurement

(i) temp. → thermocouple, RTD

(ii) Pressure → Barometre, U-tube manometre

(iii) ~~Velocimetre~~ rate of flow → Venturimetre, orifice mitre, Rotametre.

→ these are measured under steady state condⁿ.

Characteristics of Transducer:-

(Static)

(i) Accuracy. → closeness of measured value to true value.

(ii) Precision → repeatability of measurement.

(iii) Sensitivity → ratio of O/p change to the I/p change.

Linearity :- O/p should be directly proportional to I/p

Resolution :- Smallest detectable I/p change.

Drift :- Change in O/p over time without change in I/p.

Dynamic Characteristics
These are imp. when I/p changes with time.

(i) Speed of Response :- how fast O/p reaches to steady value

(ii) Lag :- Delay b/w I/p & O/p

(iii) Time constant :- time taken to reach 63.2% of final value

(iv) Bandwidth :- Range of frequencies over which sensor work accurately / properly

Comparison b/w Active & Passive Transducer

Feature	Active	Passive
(i) Power supply	Not Required	required
(ii) O/p	direct	indirect
Ex	Thermocouple, photo voltaic cell	Strain gauge, LVDT

Application of Transducer

(i) Industrial Automation
(ii) Robotics.
(iii) Aerospace System
(iv) Biomedical instruments

Cast Iron:-

It is an iron carbon alloy, containing more than about 2% carbon generally 2%-4% along with small amount of Silicon, Mn, Sulphur, phosphorus. It is produced mainly by melting pig iron in a furnace a famous coupola

It is widely used bez of:
- good castability
- good ~~machinability~~ machinability
- high compressive strength
- relative low cost

Classification of cast iron:-
- Cast iron is classified mainly based on the form in which carbon appears in it microstructure.

-) Grey cast iron:
In this type, carbon exist in the graphite ~~flex~~ flaks.

Character:-
→ Grey fracture surface due to graphite flex good
→ good machenability
→ exellent vibration damping capacity

Application:-
→ Engine blocks
→ Machine tool bodies
⇒ pump industries pipe & fittings.

(i) White Cast Iron:-
In this type, carbon exist in combine form as cementite (Fe_3C) / Iron carbide
→ highest % of carbon in iron carbide is 6.67%
↓
It is an intermetallic compound

Characteristics
→ hard & brittle
→ very high wear resistance
→ white crystalline fracture surface

Applications
→ wear resistance surface
→ Crushing rolls
→ ~~automobile~~

(ii) Malleable cast iron:-
→ produced by heat treating white cast iron, which convert cementite into temper carbon

Characteristics
→ Improved Ductility
→ Good shock resistance
→ Better machenability than white cast iron

Application.
→ Automobile industries
→ Agricultural Machinery
→ pipe fitting

(iv) Nodular cast iron
→ It is also known as ductile cast iron
In this type, graphite exist as spherical nodules instead of flakes. This is achieve by adding

Mg Magnesium or cerium during casting.

Character:
→ high Strength
→ Good Ductility
→ Good fatigue resistance

Application
→ crank shaft
→ Gears
→ pressure pipe

properties of cast iron :-

① Mechanical properties
a) high Compressive strength :- Cast iron can withstand at very high compressive loads which makes it useful for structural machine parts.
b) low tensile strength :-
Compared with steel, cast iron has lower tensile strength
c) Brittleness
most types of cast iron are little & fracture without plastic deformation.
d) Good wear resistance
Due to the presence of graphite or cementite, cast iron shows excellent wear resistance

② physical properties
(i) Good castability :-
Cast iron flows easily in molten form & fills complex mold shape.
(ii) Good thermal conductivity :-
(iii) Good vibration & damping characteristics

Chemical properties

a) Corrosion resistance
Cast iron has moderate corrossion resistance compared with plane carbon steel

b) oxidation.
Some alloy cast iron exhibit good oxidation resistance at high temp.

Advantages of cast iron:

→ Low manufacturing cost
→ Excellent machinability
→ Good wear resistance
→ Good vibration & damping characteristics
→ Good fluidity during casting

Limitation of Cast iron:-

→ brittle in nature
→ poor impact resistance
→ low tensile strength (except Ductile iron)
→ Difficult to weld

Difference feature	Grey cast iron	white cast iron
→ Form of Carbon	→ Carbon exists as graphite flex	→ Carbon exists in combine form as cementite
→ structure surface	→ Grey coloured fracture surface	→ white coloured fracture surface
→ hardness	→ relatively soft	→ Very hard
→ Machinability	→ Good	→ poor

→ Strength	→ Good compressive strength, low tensile strength	→ very high compressive strength
→ Ductility	→ slightly better than white cast iron	→ very brittle
→ wear resistance	→ moderate	→ very high.
→ shock resistance	→ Better shock absorption due to graphite	→ Very poor shock resistance
→ high conductivity	→ Good thermal conductivity	→ Lower than grey cast iron.

Difference.

Feature	Grey cast iron	Steel
→ Carbon content	→ high carbon content (2-4 %)	→ low carbon content (0-2%)
→ form of carbon	→ carbon exists as graphite flakes	→ carbon exists mainly in combine form (cementite)
→ strength	→ high tensile strength &	high tensile strength & compressive strength (But compressive lower than tensile)
→ ductility	→ Brittle, very low ductility	→ very high ductility
→ Machinability	→ Good machinability	→ moderate
→ millability	→ difficult	→ easy
→ shock resistance	→ Good vibration damping but impact resistance	→ better shock & impact resistance.

melting point	→ low	→ high
Application	→ engine blocks, pipe etc.	→ structural components shaft, gear, construction materials

High speed steel: (HSS)

→ It is a type of alloy tool steel that can cut metals at high speeds without loosing its hardness due to heat generated during machining
→ It retains hardness at high temperature around 600°C
→ HSS is mainly used for cutting tools

Composition of HSS:-

(W) Tungsten, molybdenum (Mo), chromium (Cr), vanedium (V), C

Properties

→ Very high hardness
→ high wear resistance
→ Good cutting ability
→ high strength.
→ maintain hardness at high temp.

Application

→ Drill bits
→ milling cutters
→ lathe tools
→ rimmers

Stainless steel :-

It is an iron-carbon alloy contains atleast about 10.5% to 30% chromium, which provides excellent corrosion resistance by forming a protected a oxide layer on the surface. Typical

Typical composition

(i) Chromium (Cr) → 10.5 – 30 %.
(ii) Carbon → less than 1.2%.
(iii) Ni → 0 – 20%
(iv) Mo → 0 – 7%

Properties

→ Excellent corrosion resistance
→ Good strength
→ Good oxidation resistance
→ Good ductility

Applications

→ Utensils
→ Medical instruments
→ Chemical industrial equipment
→ Structural component
→ Food processing equipment

Feature	HSS	SS
→ main purpose	→ cutting tools	→ corrosion resistance
main alloying element	→ W, Mo	→ Chromium
heat resistance	→ Very high	→ moderate.
Corrosion resistance	→ moderate	→ Very high